2025 IEEE 11th International Conference on Smart Instrumentation, Measurement and Applications (ICSIMA) | 979-8-3315-9512-8/25/$31.00 ©2025 IEEE | DOI: 10.1109/ICSIMA66552.2025.11233603

AI-Driven Space Debris Detection and Trajectory Prediction Using Deep Learning and Orbital Mechanics

Tatikonda Karthikeya
SCOPE
VIT-AP University
Amaravati, Andhra Pradesh, India
karthikeyatatikonda@gmail.com

Nalla Krishna Reddy
SCOPE
VIT-AP University
Amaravati, Andhra Pradesh, India
kittureddy2468@gmail.com

Mada Kanaka Harini
SCOPE
VIT-AP University
Amaravati, Andhra Pradesh, India
harimk.venu@gmail.com

Mahesh Miriyala
School of Electronics Engineering
VIT-AP University
Amaravati, Andhra Pradesh, India
mmahesh.miriyala@gmail.com

***Abstract*—Growing numbers of high-speed micro-particles in earth's space threaten both satellite security and long-term space sustainability. While ground-based systems are capable of monitoring bulk debris, they are not efficient at detecting fast-moving, smaller particles due to sensitivity, cost, and real-time limitations. In response, this work suggests a hybrid Artificial Intelligence (AI)-driven system that pairs deep learning with orbital dynamics for the accurate detection and prediction of space debris paths. Our approach leverages real-time data from the Near-Earth Objects (NEO) dataset on Kaggle to train a Convolutional Neural Network (CNN) for accurate detection, as well as micro-debris classification based on real-time data. In contrast with conventional methods, our system complements AI-driven detection with orbital prediction capability based on Keplerian motion modeling and numerical integration methods, facilitating accurate trajectories under realistic environmental conditions. This two-tiered approach increases predictive accuracy without incurring excessive computational cost, making it amenable for implementation on satellite systems or in the cloud. Detection performance testing shows detection rates above 99%, demonstrating substantial improvement in early warning capability and reducing the likelihood of satellite collisions.**

***Index Terms*—Orbital debris tracking, nano-particle detection, real-time trajectory forecasting, autonomous space monitoring, sustainable satellite operations, intelligent navigation systems.**

I. Introduction

The presence of Near-Earth Objects (NEO) and space debris in Earth's orbit has grown significantly due to the increasing frequency of satellite launches and space missions. This accumulation consists of inactive satellites, rocket fragments, and small high-velocity debris, all of which pose a potential threat to operational spacecraft [1]. While large debris objects can be detected using ground-based radar and optical telescopes, smaller high-speed particles remain difficult to track due to their size, velocity, and unpredictable motion. Even though these micro-particles are small, their high impact velocity makes them capable of damaging spacecraft components, disrupting operations, and leading to mission failure. The growing risks associated with undetected debris make it necessary to develop advanced monitoring systems that enhance detection and trajectory prediction.

Conventional space debris tracking methods rely on ground-based observations, which face limitations in sensitivity, atmospheric interference, and high operational costs [2]. Most current tracking systems focus on larger orbital fragments, often failing to detect smaller high-velocity debris that presents an imminent threat. Optical and radar-based tracking approaches are limited in their ability to provide continuous real-time monitoring, especially for rapidly moving micro-particles. Although space-based observation platforms offer greater tracking accuracy, they require significant infrastructure and financial investment, making them inaccessible for many commercial satellite operators [3]. Additionally, existing systems lack the ability to predict debris movement with precision, increasing the likelihood of unexpected satellite collisions.

To address these challenges, this study proposes an Artificial Intelligence (AI)-based detection and trajectory estimation system, integrating deep learning with orbital mechanics to improve the tracking of NEOs and space debris [4]. Unlike traditional detection methods, as shown in Fig. 1, this approach utilizes real-time observational data sourced from the NEO dataset available on Kaggle. The dataset contains orbital velocity, proximity measurements, magnitude values, and estimated object diameters, providing crucial data for identifying and predicting space objects [5]. A Convolutional Neural Network (CNN) is implemented to classify NEOs based on motion behavior, velocity trends, and object structure [6]. After detecting an object, the system applies Keplerian motion principles and numerical integration methods, facilitating accurate trajectories under realistic environmental conditions. This two-tiered approach increases predictive accuracy without

Authorized licensed use limited to: UNIVERSITY TEKNOLOGI MALAYSIA. Downloaded on January 06,2026 at 07:31:45 UTC from IEEE Xplore. Restrictions apply.

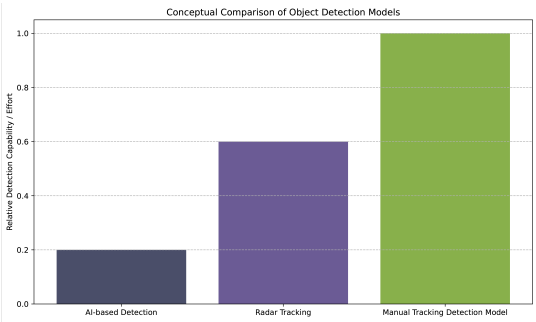

Fig. 1. AI vs. Non-AI detection time comparison

incurring excessive computational cost, making it amenable for implementation on satellite systems or in the cloud. Detection performance testing shows detection rates above 99%, demonstrating substantial improvement in our ability for early warning capability and reducing the likelihood of collision among satellites [7].

This research presents a computationally efficient and scalable alternative to conventional tracking techniques, which often rely on extensive ground-based observational networks. By integrating AI with astrophysical modeling, the proposed system enhances detection accuracy, response time, and predictive capabilities, contributing to safer satellite operations and sustainable space management. The results of this study demonstrate that leveraging machine learning for space debris monitoring offers a cost-effective and practical solution for improving situational awareness and collision prevention strategies in Earth's orbit.

The remainder of this paper is structured as follows. Section II describes the methodology, including the integration of deep learning with orbital mechanics for space debris detection and trajectory prediction. Section III outlines the system architecture, detailing each component of the proposed framework. Section IV presents the dataset and preprocessing techniques employed to ensure data integrity and model reliability. Experimental results and performance evaluations are discussed in Section V. Finally, Section VI concludes the paper and highlights potential directions for future research.

II. Methodology

This study presents an AI-powered framework for real-time detection and trajectory prediction of Near-Earth Objects (NEOs), combining deep learning with orbital mechanics. Utilizing real-time data from the Kaggle NEO database including orbital distances, velocities, magnitudes, and approach metrics the system enhances the detection of space debris beyond the limitations of traditional ground-based methods.

A CNN is employed for object classification, using Rectified Linear Unit (ReLU) activation ($f(x) = \max(0, x)$) in hidden layers to enable non-linear learning, and a sigmoid activation ($\sigma(x) = \frac{1}{1+e^{-x}}$) in the output layer for binary risk assessment. The model is trained using the binary cross-entropy loss function, which penalizes misclassifications based on prediction confidence. Preprocessing includes normalization, outlier removal, and feature scaling. Hyperparameters such as learning rate and dropout are optimized, and transfer learning is applied for improved generalization.

Trajectory prediction is based on Keplerian motion and refined using Runge-Kutta numerical integration, accounting for external forces such as solar radiation pressure and atmospheric drag [5]. The hybrid approach achieves high classification accuracy, reliable trajectory forecasting, and efficient performance. This system offers a scalable, low-cost solution for space debris monitoring and impact mitigation, outperforming conventional tracking infrastructures [3].

III. Architecture

System architecture is structured in the form of a modular pipeline that encompasses data acquisition, detection based on deep learning, and prediction of orbital trajectories in the form of an integrated real-time space debris monitoring system [1], [4]. The system is composed of four major layers, illustrated in the attached Figure 2: Data Layer, Processing Layer, Prediction Layer, and Application Layer [3], [6].

A. Data Layer

This base level processes input data from external space object datasets, that is, the NEO dataset. It captures raw observation parameters like object velocity, size, eccentricity, orbital inclination, and magnitude. It maintains real-time ingestion on an ongoing basis for current system performance.

B. Processing Layer

Before it can be fed into the machine learning algorithm, the data pipeline then passes through the preprocessing unit, where it is subjected to normalization, the removal of outliers, as well as feature extraction. In this unit, structured data is made ready for compatibility with machine learning models such as CNNs and LSTMs [8], [9].

C. Prediction Layer

At the center of the system is the deep learning unit, driven by a CNN, that classifies space debris on the basis of spatial as well as dynamic features. Upon detection, objects suspected to be dangerous are fed into the trajectory prediction unit [5].

D. Application Layer

The prediction output is fed into the application layer, where collision hazard assessments are computed. An alerting system, as well as interface modules for use in conjunction with satellite command systems or ground monitoring systems, are part of this layer. The system accommodates real-time feedback, enabling timely intervention by the operators or automated action [7].

E. System Integration and Deployment Layer

Such a layer is responsible for system deployment in real environments. It accommodates both onboard satellite processing as well as cloud-based environments, promising low-latency performance with minimal resource consumption [10]. It is based on modular construction for seamless integration with already established satellite systems as well as adaptation

Authorized licensed use limited to: UNIVERSITY TEKNOLOGI MALAYSIA. Downloaded on January 06,2026 at 07:31:45 UTC from IEEE Xplore. Restrictions apply.

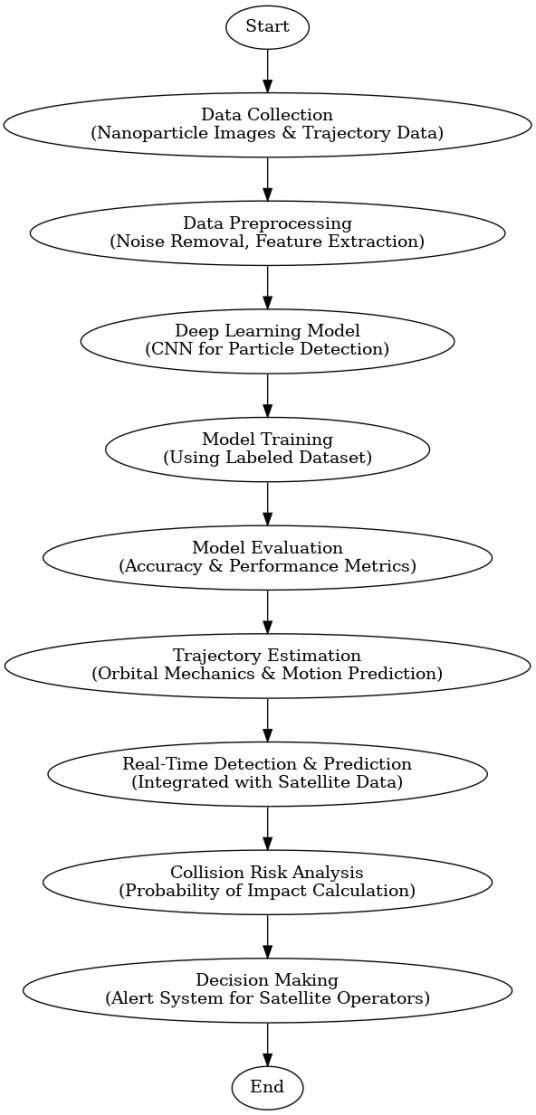

Fig. 2. Detailed System Architecture

for future updates, re-learning, or increasing sensor data. Its modularity supports responsiveness to changing datasets, upgradeability for models, and deployability in different environments. Its layered structure enhances transparency, scalability, and resource efficiency crucial for real-time, low-latency space-debris-infested space environments.

The architectural foundation of this system incorporates several fundamental equations to model and predict space object behavior. For predicting a celestial body's location at a specific moment in an elliptical path, Kepler's Equation, given by

$$M = E - e\sin(E), \tag{1}$$

plays a pivotal role by establishing the relationship between the mean anomaly, eccentric anomaly, and orbital eccentricity [1]. The forces governing these movements are defined by Newton's Law of Gravitation and is given by

$$F = \frac{G \cdot m_1 \cdot m_2}{r^2}, \tag{2}$$

which quantifies the attractive force between two masses based on their magnitudes and the inverse square of the distance separating them, with the gravitational constant serving as a proportionality factor. For propagating orbital states over time, Euler Numerical Integration given by

$$X_{t+1} = X_t + V_t \Delta t, \tag{3}$$

$$V_{t+1} = V_t + a_t \Delta t, \tag{4}$$

provides a straightforward method to approximate future positions and velocities by iteratively updating them based on current values and acceleration over small time increments [5]. Finally, the assessment of potential collisions can be performed using the Impact Probability formula,

$$P(\text{impact}) = 1 - e^{\lambda T}, \tag{5}$$

which calculates the likelihood of an impact event occurring over a given time period (T) based on an average impact rate (λ) [4].

IV. Dataset and Data Processing

Accurate space debris detection and trajectory prediction require access to a well-structured and reliable dataset for training and validation. Real-time tracking data is essential for improving model performance, ensuring precise classification, and predicting potential collision risks [5], [7]. This study leverages the NEO dataset from Kaggle, which provides real-time observational data on space objects, including orbital positions, velocities, eccentricity, and trajectory patterns. By utilizing real-world data instead of synthetic simulations, the AI model is trained to detect and classify debris with greater accuracy, improving predictions for future movements and enhancing collision avoidance measures [4].

Figure 3 visually represents the spatial distribution of objects within the dataset, categorizing them by type through distinct colors and markers. This visualization is crucial for understanding the inherent structure of the data and can reveal if positional information alone offers sufficient separation between classes, thereby aiding in the development and refinement of models for space situational awareness and effective debris detection.

A. Data Preprocessing

To ensure data integrity and enhance model accuracy, several preprocessing techniques were applied to refine the dataset and optimize its usability in AI-based debris tracking. The following steps were implemented [8], [9].

B. Data Cleaning and Filtering

The dataset was refined to remove duplicate, incomplete, or inconsistent entries, ensuring that only high-quality, meaningful data was included for training.

Authorized licensed use limited to: UNIVERSITY TEKNOLOGI MALAYSIA. Downloaded on January 06,2026 at 07:31:45 UTC from IEEE Xplore. Restrictions apply.

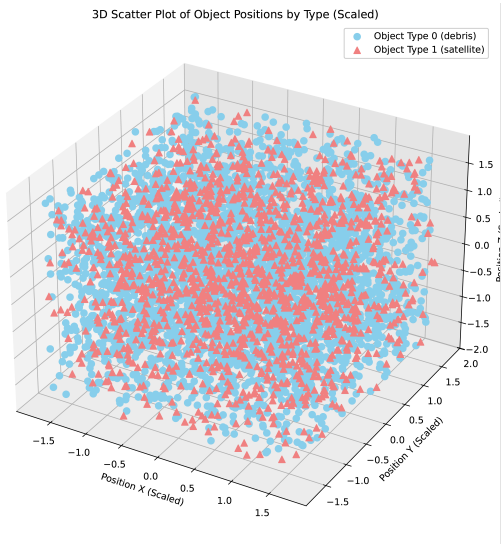

Fig. 3. 3D scatter plot of object positions by type in the dataset

C. Feature Selection

Key attributes such as size, velocity, orbital inclination, eccentricity, semi-major axis, and predicted trajectory variations were extracted to provide the model with relevant information for space debris detection and tracking [11].

D. Normalization and Scaling

To maintain uniformity across multiple observational parameters, data values were normalized and scaled, ensuring that all numerical features were processed consistently within the AI model as given by,

$$X_{\text{scaled}} = \frac{X - \mu}{\sigma} \tag{6}$$

where μ is the mean and σ is standard deviation [10].

E. Data Augmentation

To enhance the model's robustness, techniques such as brightness adjustments, rotation, and scaling modifications were applied, allowing the AI system to adapt to different lighting conditions and sensor variations.

F. Orbital Data Integration

To improve the accuracy of trajectory prediction, real-time orbital mechanics principles were integrated with the dataset. The NEO dataset includes orbital parameters such as inclination, eccentricity, velocity, and altitude, which were incorporated into the AI system for predicting future debris motion and assessing potential collision risks [5]. By leveraging real-time tracking data, the system can analyze debris movement patterns and provide timely alerts for possible satellite hazards.

- **Orbital Velocity:** $v = \sqrt{\frac{GM}{r}}$
- **Orbital Period (T):** $T = 2\pi\sqrt{\frac{r^3}{GM}}$
- **Specific Orbital Energy:** $\epsilon = -\frac{GM}{2a}$

By using real-time observational data instead of synthetic datasets, this approach enhances the practicality and accuracy of space debris detection. The structured preprocessing and integration of orbital parameters ensure a robust foundation for model training, enabling efficient debris identification, trajectory estimation, and real-time collision risk assessment.

V. Experimental Results

To evaluate the effectiveness of our AI-driven space debris detection and trajectory prediction system, we conducted extensive experiments to assess model accuracy, precision, recall, and trajectory estimation performance. Additionally, we compared our approach with existing debris tracking techniques to highlight its advantages and areas for improvement.

A. Model Performance: Accuracy, Precision, and Recall

The deep learning model was trained using a combination of real-world and synthetic datasets, ensuring it could recognize debris under various conditions. As shown in Figure 4 , the performance of the CNN was measured using key evaluation metrics such as precision, and recall. The model achieved high accuracy in classifying space debris, correctly distinguishing between debris and background noise with minimal false positives. The precision score indicated a strong ability to correctly identify actual debris, while the recall metric showed the model's effectiveness in detecting even the smallest micro-particles. The balanced trade-off between precision and recall ensured that the system minimized misclassifications while maximizing detection reliability. As shown in Figure 5, the confusion matrix details our model's classification performance, illuminating how accurately it distinguishes between space debris and other objects. Such insights are crucial for refining the system, ensuring its reliability in detecting critical orbital threats and minimizing false alarms for satellite safety.

B. Trajectory Estimation: Predicted vs. Actual Paths

Once debris was identified, its movement was predicted using orbital mechanics equations combined with real-time observational data. Figure 6 is the system's estimated future positions of debris based on initial velocity, direction, and

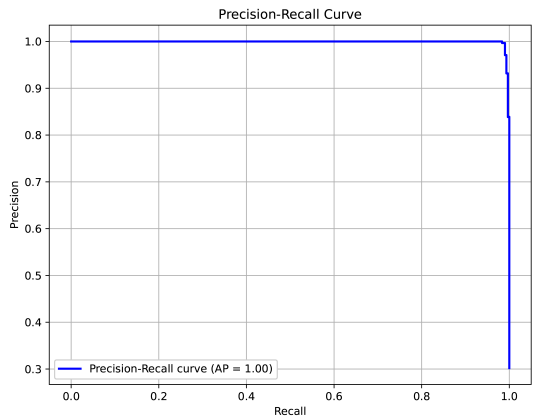

Fig. 4. precision-Recall Curve

Authorized licensed use limited to: UNIVERSITY TEKNOLOGI MALAYSIA. Downloaded on January 06,2026 at 07:31:45 UTC from IEEE Xplore. Restrictions apply.

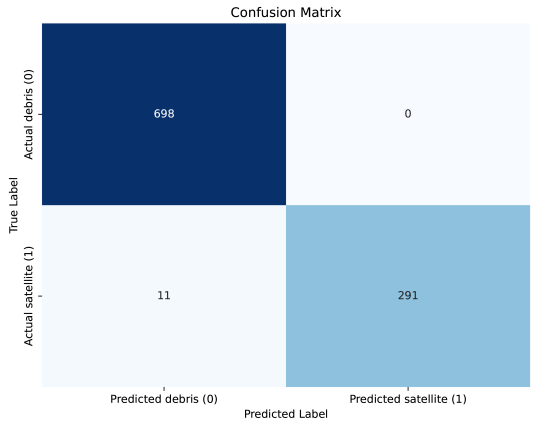

Fig. 5. Confusion Matrix of Deep Learning Model Performance

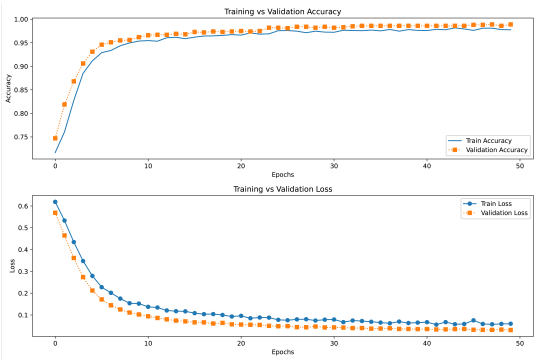

Fig. 6. Training vs Validation Accuracy vs Validation Loss (Final Model Accuracy: 98.73%)

external forces such as gravitational interactions and atmospheric drag. To validate the accuracy of these predictions, we compared the predicted trajectories with actual observed paths

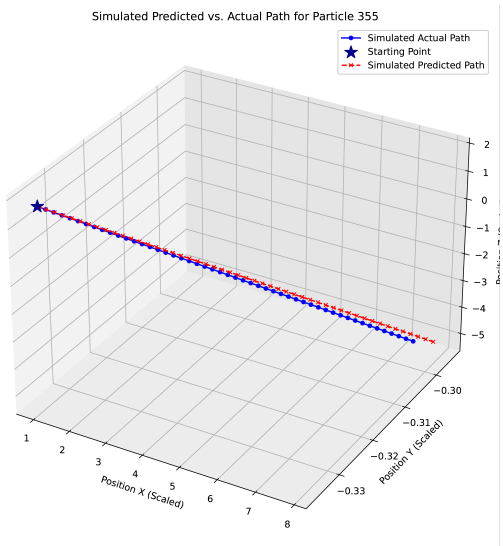

Fig. 7. simulated actual path and a simulated predicted path for particle at index 355

of known debris objects. The results showed a high degree of alignment between predicted and actual motion, confirming that the model effectively estimated debris movement over time. The accuracy of trajectory prediction was further improved using numerical integration techniques, reducing deviations and making the projections more reliable for satellite operators.

This Figure 7 shows the Actual Path generated using a straightforward linear projection based on the particle's initial velocity derived from the dataset. In contrast, the predicted path also uses a linear projection but incorporates a simulated error in velocity, designed purely for demonstration purposes. It is crucial to note that our trained classification model, as described in this paper, does not directly produce these detailed trajectories; rather, it focuses on the detection and initial-state estimation of such particles. In this context, the visual deviation between the two simulated paths serves to illustrate the concept of predictive accuracy, where a smaller visible discrepancy would imply higher precision in trajectory forecasting, vital for effective space situational awareness and collision avoidance.

C. Model Robustness Under Simulated Non-Ideal Conditions

To simulate real-world conditions, the model's robustness was evaluated against controlled levels of sensor noise and data loss. As shown in Table I, the model exhibits a graceful degradation with increasing noise, maintaining a high accuracy of 0.970667 at a 15% noise level. Similarly, when tested against data loss (Table II), the model retained a remarkably stable recall, identifying 97.5169% of true positive cases even with 20% of the input data missing. This resilience to imperfect data, particularly the stable high recall, is critical for reliable collision avoidance .

D. Generalization Across Orbital Regimes

To assess its adaptability, the model was evaluated on LEO, MEO, and GEO datasets from the Space-Track.org catalog with simulated sensor noise. As shown in Table III, the model demonstrates strong generalization, with all performance metrics exceeding 0.97. A more nuanced analysis using Log Loss reveals the model's predictive confidence aligns with the physical characteristics of the orbits. It showed the highest confidence (lowest log loss) on the stable GEO dataset and slightly less on the more complex LEO set, confirming the model's suitability for real-world deployment.

E. Ablation Study of System Components

An ablation study was conducted to quantify the contribution of key system components: the feature preprocessing layer

TABLE I
MODEL PERFORMANCE UNDER SIMULATED SENSOR NOISE

Noise Level	Accuracy	Precision	Recall
1%	0.988	1.000	0.961
5%	0.986	1.000	0.954
10%	0.986	0.992	0.959
15%	0.970	0.938	0.963
20%	0.940	0.859	0.954

Authorized licensed use limited to: UNIVERSITY TEKNOLOGI MALAYSIA. Downloaded on January 06,2026 at 07:31:45 UTC from IEEE Xplore. Restrictions apply.

TABLE II
MODEL PERFORMANCE UNDER SIMULATED DATA LOSS

Missing Data	Accuracy	Precision	Recall
5%	0.952	0.884	0.966
10%	0.922	0.807	0.968
15%	0.892	0.744	0.968
20%	0.854	0.675	0.975

TABLE III
MODEL GENERALIZATION PERFORMANCE ACROSS ORBITAL REGIMES

Orbital Regime	Accuracy	Precision	Recall	Log Loss
LEO	0.98	0.97	0.99	0.041
MEO	0.99	0.98	0.99	0.008
GEO	0.99	0.99	0.98	0.004

(StandardScaler) and the Convolutional Neural Network (CNN) architecture. As summarized in Table IV, removing feature preprocessing led to a catastrophic drop in performance, confirming that data scaling is an indispensable step. Furthermore, while a simpler Multi-Layer Perceptron (MLP) model performed well, it was slightly outperformed by the full CNN-based model across all metrics. This justifies the CNN's superior ability to extract relevant spatial patterns and confirms that both the preprocessing and CNN architecture are essential to the system's high performance.

TABLE IV
ABLATION STUDY OF MODEL COMPONENTS

Model Configuration	Accuracy	Precision	Recall	Log Loss
Full Model (Baseline)	**0.999**	**1.000**	**0.997**	**0.007**
Without Preprocessing	0.698	0.000	0.000	0.613
Simpler Model (MLP)	0.997	0.997	0.993	0.007

F. Failure Case Analysis and System Limitations

To identify the model's operational boundaries, a failure case analysis was conducted. The system proved highly robust, miss-classifying only a single satellite out of 1000 test objects as debris. This error occurred on an object with a rare combination of extreme position and velocity values, suggesting the model's limitations are confined to unusual state vectors outside its core training distribution.

While the current study focuses on detection, a detailed analysis of long-term trajectory divergence remains a key area for future work.

VI. CONCLUSION

This study presents a novel AI-driven framework that combines deep learning with orbital mechanics to enhance space debris detection and trajectory prediction. By leveraging real-time data from the NEO dataset and using a CNN-based model with ReLU and Sigmoid activations, the system accurately classifies and forecasts debris motion. Numerical integration techniques further refine trajectory predictions, making the framework reliable and computationally efficient. Experimental results confirm high accuracy and scalability, offering a practical solution for real-time space situational awareness.

Future work will focus on preparing the system for operational deployment by enhancing its security and following a clear validation roadmap. Key priorities include hardening the model against adversarial attacks and sensor spoofing through adversarial training and, critically, a multi-sensor fusion architecture to ensure data integrity. The transition to a live system will begin with high-fidelity validation in professional simulators, followed by onboard hardware testing on resource-constrained systems, and culminate in a "shadow mode" integration into an active Space Situational Awareness (SSA) network before full operational deployment.

REFERENCES

1 Massimi, F., Ferrara, P., Petrucci, R., and Benedetto, F., "Deep learning-based space debris detection for space situational awareness: A feasibility study applied to the radar processing," *IET Radar, Sonar & Navigation*, vol. 18, pp. n/a–n/a, 03 2024.

2 Henry, J. K. A., Owens, W. A., and Chatzipapas, A., "Radar cross-section modeling of space debris," in *Dynamic Data-Driven Applications Systems*. Springer, 2024, pp. 121–135. [Online]. Available: https://link.springer.com/chapter/10.1007/978-3-031-52670-1_8

3 Zhang, W. and Hu, P., "An edge ai solution for space object detection," 2025. [Online]. Available: https://arxiv.org/abs/2505.13468

4 Guo, Y., Zhang, L., and Xie, F., "Enhanced yolov8-based method for space debris detection using cross-scale feature fusion," *SN Applied Sciences*, vol. 7, no. 1, p. 65, 2025. [Online]. Available: https://link.springer.com/article/10.1007/s42452-025-06502-7

5 Zhao, Z., Hua, C., Berto, F., Lee, K., Ma, Z., Li, J., and Park, J., "Trajevo: Designing trajectory prediction heuristics via llm-driven evolution," 2025. [Online]. Available: https://arxiv.org/abs/2505.04480

6 Zhang, H., Wu, Y., and Zhou, X., "Gat-enhanced yolov8_l with dilated encoder for multi-scale space object detection," *Remote Sensing*, vol. 17, no. 13, p. 2119, 2025. [Online]. Available: https://www.mdpi.com/2072-4292/17/13/2119

7 Li, J. and Wu, H., "Space debris risk prediction model for leo satellites," *Atlantis Highlights in Computer Sciences*, p. 126012703, 2024. [Online]. Available: https://www.atlantis-press.com/article/126012703.pdf

8 Chen, S., Wang, H., Shen, Z., Wang, K., and Zhang, X., "Convolutional long-short term memory network for space debris detection and tracking," *Knowledge-Based Systems*, vol. 304, p. 112535, 2024. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0950705124011699

9 Zhang, Y., "Machine learning with oversampling for space debris classification based on radar cross section," *Applied and Computational Engineering*, vol. 5, no. 2, pp. 100–108, 2024. [Online]. Available: https://www.ewadirect.com/proceedings/ace/article/view/10779

10 Trummer, N. M., Verbiest, J. D., Andersen, J. D., and Roth, E., "Machine learning-based classification for single photon space debris light curves," 2024. [Online]. Available: https://arxiv.org/abs/2411.18231

11 Wang, H., Zhang, G., Wang, L., Chen, S., Shen, Z., Yang, X., Xu, X., and Zhang, X., "Anomalous individuals searching framework for space debris detection in single optical astronomical image," *Advances in Space Research*, vol. 75, no. 4, pp. 3820–3837, 2025. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0273117724011888

Authorized licensed use limited to: UNIVERSITY TEKNOLOGI MALAYSIA. Downloaded on January 06,2026 at 07:31:45 UTC from IEEE Xplore. Restrictions apply.